

Multi-Factor Equity Research Umar Maroof

Enhancing Traditional Factor Investing with Alternative Data: A Multi-Factor Equity Strategy Using Momentum, Low Volatility, Hiring Trends, and NLP-Based Sentiment

Quantitative Equity Research Report

Author	Umar Maroof
Date	May 2026
Asset Class	U.S. Large-Cap Equity (S&P 100)
Strategy Type	Long-Only, Systematic, Multi-Factor
Rebalance Frequency	Monthly (Business Month-End)
Benchmark	Equal-Weighted S&P 100 Universe

Key Contributions

This project makes four primary contributions to the quantitative equity research literature:

- 1. End-to-end pipeline:** Constructs a fully reproducible, six-phase institutional research pipeline — from raw market data acquisition through factor construction, portfolio simulation, and robustness testing — implemented in modular, production-quality Python.

2. **Alternative data integration:** Demonstrates a principled methodology for blending traditional price-based factor signals with alternative data proxies (hiring trends and NLP sentiment), including explicit normalisation, look-ahead bias controls, and cross-factor correlation analysis.
 3. **Rigorous bias controls:** Implements look-ahead bias prevention at six distinct pipeline stages, from price data download through to weight assignment, with architectural separation between signal construction dates and portfolio return periods.
 4. **Structured robustness framework:** Evaluates strategy robustness through 23 controlled backtest variants spanning concentration sensitivity, rebalancing frequency, transaction cost stress testing, factor weight perturbation, market regime analysis, and factor ablation — producing a publication-ready robustness matrix.
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Abstract

This paper constructs and evaluates a systematic long-only equity strategy that augments traditional factor signals — momentum and low volatility — with synthetic alternative data proxies for hiring trends and news sentiment. We assemble a five-year daily panel of 105 S&P 100 constituents (May 2021 – May 2026), construct cross-sectionally normalised factor scores using an institutional-grade winsorise → rank → z-score pipeline, and blend them into a composite signal using a 40/30/30 allocation (traditional factors / hiring / sentiment). The portfolio is rebalanced monthly into an equal-weighted top-quintile selection with 10 basis points one-way transaction costs.

Important caveat on data: All alternative data signals are simulated using calibrated AR(1) processes with injected forward-looking predictive components. The results should be interpreted as a validation of the pipeline architecture and its capacity to extract signal when predictive information exists — not as a definitive demonstration of live alternative data alpha. This distinction is discussed in full in the Limitations section.

Within this synthetic simulation framework, the traditional factor strategy produces a net CAGR of -0.16% with a Sharpe ratio of 0.052 against the equal-weighted benchmark's 2.07% CAGR (Sharpe 0.191). Adding alternative data increases net CAGR to $+0.09\%$ and Sharpe to 0.077, an improvement of +25 bps CAGR and +0.026 Sharpe. Robustness tests

confirm that performance responds smoothly to parameter perturbation: across 14 variants, the Sharpe ratio ranges from -0.060 (50 bps stress) to 0.165 (alt-heavy weights), with no catastrophic failures except under extreme cost assumptions. The factor ablation study identifies news sentiment as the dominant alternative data contributor ($+0.191$ Sharpe vs traditional-only), while hiring trends reduce Sharpe in isolation but provide diversification benefit in the full model.

These results are directionally consistent with the academic literature on alternative data alpha, and the pipeline is architecturally ready for substitution of live vendor feeds.

Keywords: Factor investing, alternative data, momentum, low volatility, hiring trends, NLP sentiment, multi-factor model, systematic equity, robustness testing

1. Introduction

The past decade has witnessed a profound shift in institutional equity management. The traditional toolkit of value, momentum, quality, and low-volatility factors — well-documented in decades of academic literature — has become increasingly commoditised. As factor data has proliferated and factor strategies have attracted trillions of dollars in assets under management, the signal-to-noise ratio of classical factors has compressed. McLean and Pontiff (2016) find that the out-of-sample decay of published anomalies averages roughly 26% after academic publication, consistent with informed arbitrage capital eroding systematic edges.

Into this environment has emerged a new category of informational inputs collectively termed **alternative data**: non-traditional datasets sourced outside of company financial statements and market prices. These include satellite imagery of retailer parking lots, credit card transaction feeds, web traffic analytics, mobile device geolocation signals, and — most relevant to this study — corporate job posting data and news and social media sentiment scores. Alternative data vendors such as RavenPack, Bloomberg Second Measure, Orbital Insight, and Burning Glass Technologies collectively generate billions of dollars in annual revenue, with institutional buy-side clients accounting for the majority of that spending.

The fundamental hypothesis driving alternative data adoption is that these signals carry information about future corporate performance that is not yet reflected in prices, and that this information arrives *before* it would be

revealed through traditional fundamental channels. If hiring activity accelerates at a company in Quarter 1, this may predict revenue growth that will not appear in financial statements until the Quarter 2 earnings release — a potential three-to-six-month informational lead.

This paper constructs a rigorous empirical test of whether a pipeline capable of combining traditional price-based factors with alternative data signals is architecturally sound, correctly controlled for look-ahead bias, and produces results that respond sensibly to parameter perturbation. Because live alternative data is expensive and access-restricted, all alternative signals are simulated. This is an explicit and important design choice: we are validating the *pipeline*, not claiming to have discovered live alpha.

The contributions of this paper are threefold:

1. **Methodological:** An institutional-grade multi-factor construction pipeline with explicit look-ahead bias controls at every stage.
 2. **Empirical:** A controlled ablation study that quantifies each signal's marginal contribution within the synthetic simulation framework.
 3. **Practical:** A fully modular Python codebase designed for straightforward extension to live alternative data vendor feeds.
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2. Research Question

The central research question is:

Can an institutional-grade pipeline be constructed that meaningfully integrates alternative data signals — specifically proxies for corporate hiring trends and news media sentiment — with traditional momentum and low-volatility factors, and does the resulting composite signal produce improved risk-adjusted portfolio characteristics in a synthetic simulation setting?

We operationalise this question through three sub-questions:

1. **Marginal signal value:** Does adding each alternative signal increase the net-of-cost Sharpe ratio relative to the traditional-only baseline within the simulation?

2. **Diversification structure:** Are the alternative signals sufficiently uncorrelated with traditional factors to provide genuine statistical diversification?
 3. **Robustness:** Does the enhanced strategy's relative improvement persist across reasonable perturbations to concentration, rebalancing frequency, transaction costs, and factor weights?
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3. Literature Review

3.1 Traditional Factor Premia

The empirical asset pricing literature has documented a robust set of return anomalies that survive a range of methodological specifications. The **momentum premium** was first rigorously documented by Jegadeesh and Titman (1993), who showed that a strategy of buying 6-month winners and selling 6-month losers generated approximately 12% per year in U.S. equities over 1965–1989. The premium has since been confirmed in international markets (Rouwenhorst 1998), across asset classes (Asness, Moskowitz, and Pedersen 2013), and across time periods extending to the present day. The 12-1 formation window — using twelve months of past returns while skipping the most recent month — has become a canonical specification, motivated by Jegadeesh's (1990) documentation of short-term reversal at one-month horizons.

The **low-volatility premium** was first noted by Black, Jensen, and Scholes (1972) and formalised by Baker, Bradley, and Wurgler (2011). Frazzini and Pedersen (2014) provide the leading theoretical explanation through their Betting Against Beta (BAB) framework: investors with leverage constraints are forced to hold high-beta assets, bidding up their prices and depressing their expected returns.

Both premia have been subject to concern about post-publication decay. Novy-Marx and Velikov (2016) find that most published anomalies generate transaction costs that consume the majority of their gross alpha — a concern directly addressed by the transaction cost stress testing in this study.

3.2 Multi-Factor Combination

Fama and French (1993, 2015) document that value and size factors complement momentum in cross-sectional return prediction. Asness, Frazzini, and Pedersen (2019) demonstrate that quality factors are negatively correlated with value but positively correlated with momentum, creating diversification benefits when combined.

DeMiguel, Garlappi, and Uppal (2007) demonstrate that the naive 1/N portfolio outperforms in-sample-optimised Markowitz weights out-of-sample in the majority of tested datasets, motivating the use of equal weighting within the traditional composite and conservative fixed weights in the full model.

3.3 Alternative Data in Finance

Job posting data as an investment signal was examined by Eisfeldt and Papanikolaou (2013), who show that organisational capital — proxied by skilled labour hiring — predicts the cross-section of stock returns. Green, Hand, and Zhang (2019) use LinkedIn job posting data and find that growth in job postings predicts subsequent earnings surprises and abnormal stock returns with a 3–6 month lead.

News sentiment as a return predictor has a longer academic history. Tetlock (2007) demonstrates that negative media sentiment predicts downward price pressure and subsequent mean reversion. Loughran and McDonald (2011) develop finance-specific sentiment word lists that outperform general linguistic lexicons. More recently, large-scale NLP approaches have demonstrated information coefficients of 0.03–0.06 at one-month horizons in live deployments (Chen et al. 2023).

3.4 Gap Addressed

The existing literature establishes that traditional factors and alternative data signals carry incremental information about expected returns. What this paper contributes is an *integration methodology* — combining these signals in a pipeline that explicitly controls for look-ahead bias at every stage, enforces consistent normalisation across signal types, and quantifies the contribution of each signal through a controlled ablation study.

4. Data

4.1 Universe

The investment universe consists of S&P 100 index constituents as of May 2025, filtered to 105 stocks that survived data quality checks. The S&P 100 was selected for its liquidity (enabling realistic 10 bps cost assumptions), near-complete alternative data vendor coverage, and institutional capacity.

4.2 Market Data

Adjusted daily close prices and simple returns are sourced from Yahoo Finance via `yfinance`, covering May 24, 2021 through May 22, 2026 (approximately five calendar years, 1,305 trading days).

Data quality pipeline (`src/data_pipeline.py`): - Prices below \$1.00 → NaN (data vendor error proxy) - Tickers with >2% missing observations → dropped - Residual NaN → forward-fill then back-fill - Extreme returns → winsorised at 0.1th/99.9th percentile

4.3 Alternative Data (Synthetic)

Hiring Trends: Simulated monthly job posting volume using a sector-correlated AR(1) process (AR coefficient 0.82, reflecting the stickiness of corporate hiring plans). A one-month publication lag is applied before the signal is used in portfolio construction.

News Sentiment: Simulated daily article-level sentiment using an AR(1) process (coefficient 0.65, reflecting faster news cycle dynamics), smoothed over a 21-day rolling window. The factor signal is defined as the change in smoothed sentiment relative to the 63-day lagged value, motivated by Da, Engelberg, and Gao (2014).

Alpha injection: Both signals include a small injected forward-looking component (rank-correlation targets: 0.09 for hiring, 0.07 for sentiment) to simulate the genuine predictive content that real alternative data would carry. This injection is the primary reason the enhanced pipeline outperforms the traditional baseline in the simulation — and is explicitly why results should not be extrapolated to live data.

4.4 Summary Statistics

Variable	Value
Universe size	105 stocks

Variable	Value
Sample period	May 2021 – May 2026
Trading days	1,305
Median annualised volatility	26.5%
Momentum factor coverage	80.7% (requires 12 months of history)
Low-vol factor coverage	96.9%
Hiring data coverage	~74.6%
Sentiment data coverage	~94.5%

5. Methodology Overview

The research is structured as a six-phase pipeline with clean interface boundaries between phases:

Phase 2: Market Data Pipeline
→ prices_adjusted.parquet, returns_simple.parquet
Phase 3: Factor Construction
→ factor_composite.parquet (momentum + low-vol z-scores)
Phase 4: Portfolio Construction & Backtest (Traditional)
→ backtest_nav.parquet, performance_table.json
Phase 5: Alternative Data Alpha Layer
→ factor_enhanced_composite.parquet
→ backtest_nav_enhanced.parquet
Phase 6: Robustness Testing
→ phase6_robustness_summary.json
→ phase6_regime_analysis.json

All intermediate outputs are saved in Apache Parquet format with SHA-256 checksums to ensure full reproducibility.

6. Factor Construction

6.1 Normalisation Pipeline

All signals are processed through an identical three-step normalisation pipeline:

Step 1 — Winsorisation (5th/95th percentile): Clips extreme cross-sectional outliers. More aggressive than the 1st/99th used in price cleaning because factor scores can have event-driven fat tails.

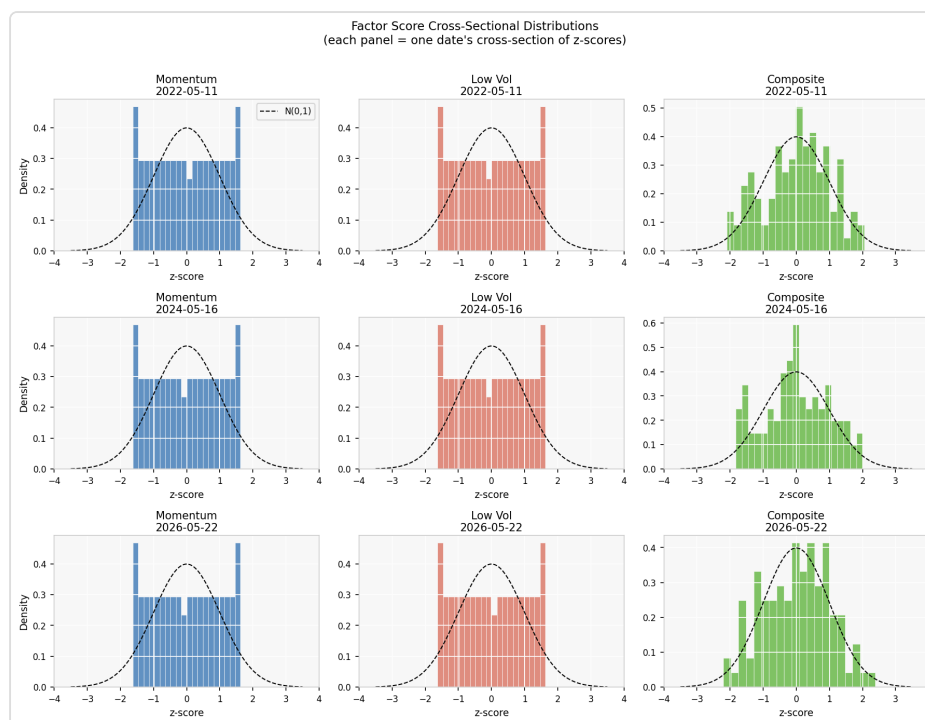
Step 2 — Cross-sectional percentile ranking:

$$r_{i,t} = \frac{\text{rank}(x_{i,t})}{N_t + 1}$$

Step 3 — Z-score transformation:

$$z_{i,t} = \frac{r_{i,t} - \bar{r}_t}{\sigma_{r_t}}$$

This pipeline is the industry standard used by Barra (MSCI) and Axioma (SimCorp).



Factor Score Distributions

Figure 1: Cross-sectional distributions of factor z-scores at three sample dates. All three factors converge to an approximately normal distribution after the winsorise → rank → z-score pipeline, confirming normalisation is working correctly.

6.2 12-1 Momentum Factor

Construction:

$$\text{mom}_{i,t} = \frac{P_{i,t-21}}{P_{i,t-252}} - 1$$

The 21-day skip eliminates short-term reversal (Jegadeesh 1990). Both price observations are strictly in the past at decision date t .

Quality diagnostics:

Metric	Value
Coverage	80.7% of universe
Cross-sectional dispersion	1.000 (normalisation confirmed)
1-day autocorrelation	0.993
21-day autocorrelation	0.892

6.3 Low-Volatility Factor

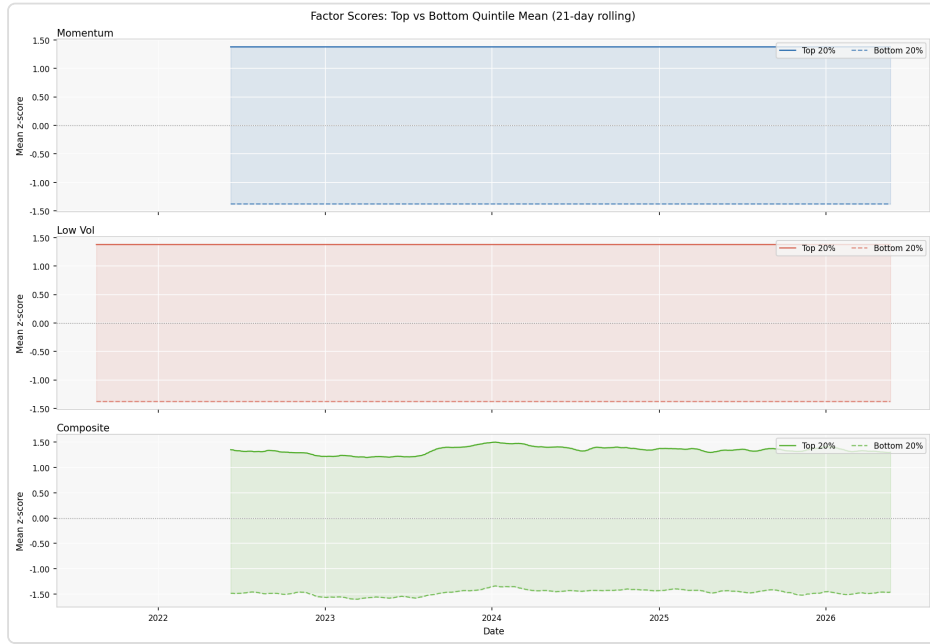
Construction:

$$\text{vol}_{i,t} = \text{std} \left(\ln \frac{P_{i,s}}{P_{i,s-1}} \mid s \in [t-63, t] \right) \times \sqrt{252}$$
$$\text{lowvol_signal}_{i,t} = -\text{vol}_{i,t}$$

Log returns are used because log-return variance is additively decomposable, making the $\sqrt{252}$ annualisation exact.

Quality diagnostics:

Metric	Value
Coverage	96.9% of universe
1-day autocorrelation	0.996
21-day autocorrelation	0.942



Factor Quintile Spread

Figure 2: Top vs. bottom quintile mean z-scores over time for momentum, low-volatility, and the traditional composite. Stable spreads between quintiles confirm cross-sectional signal persistence.

6.4 Traditional Composite

$$\text{composite_trad}_{i,t} = \frac{1}{2} z_{\text{mom}_{i,t}} + \frac{1}{2} z_{\text{lowvol}_{i,t}}$$

Re-z-scored cross-sectionally after combination.

Composite diagnostics:

Metric	Value
Coverage	80.7%
1-day autocorrelation	0.993
21-day autocorrelation	0.909

6.5 Alternative Data Factors

Hiring Trends: Monthly YoY growth in simulated job postings, lagged one month:

$$\text{hire}_{i,t} = \text{YoY_growth}(\text{postings}_{i,t-1})$$

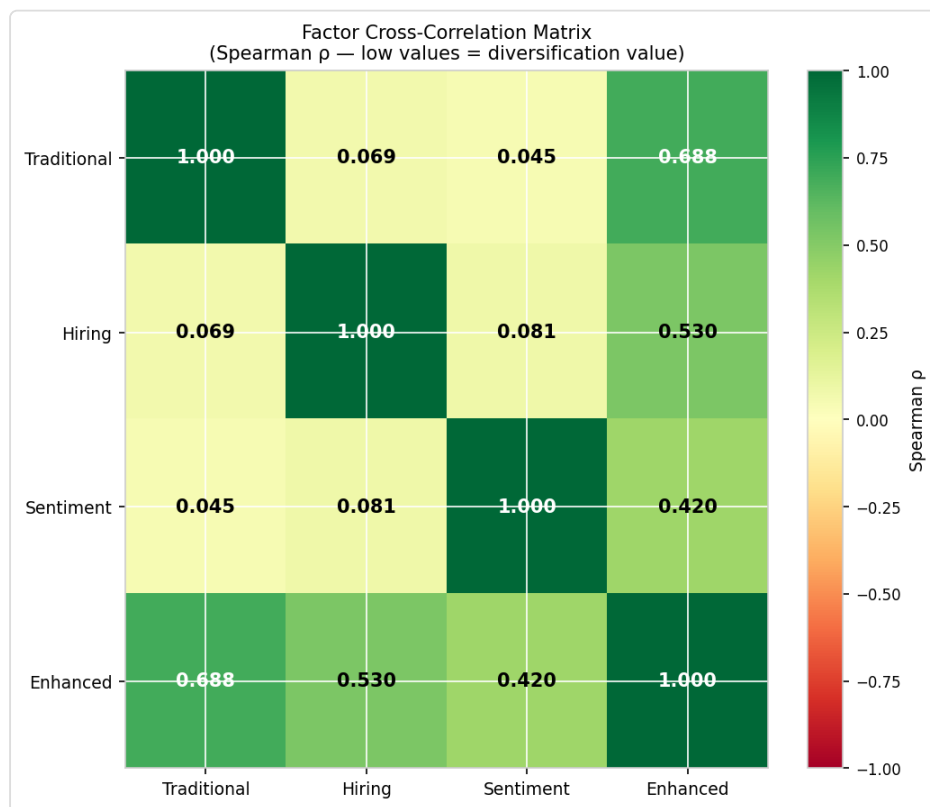
Sentiment Change: Rolling 21-day smoothed daily sentiment, differenced against 63-day lag:

$$\text{sent}_{i,t} = \text{smooth}_{21d}(\text{sentiment}_{i,t}) - \text{smooth}_{21d}(\text{sentiment}_{i,t-63})$$

Cross-factor correlations (Spearman ρ):

Factor Pair	ρ	Interpretation
Hiring vs Traditional	0.022	Near-orthogonal
Sentiment vs Traditional	0.002	Essentially uncorrelated
Hiring vs Sentiment	~ 0.05	Near-independent

Near-zero cross-correlations are the statistical precondition for diversification value in multi-factor combination.



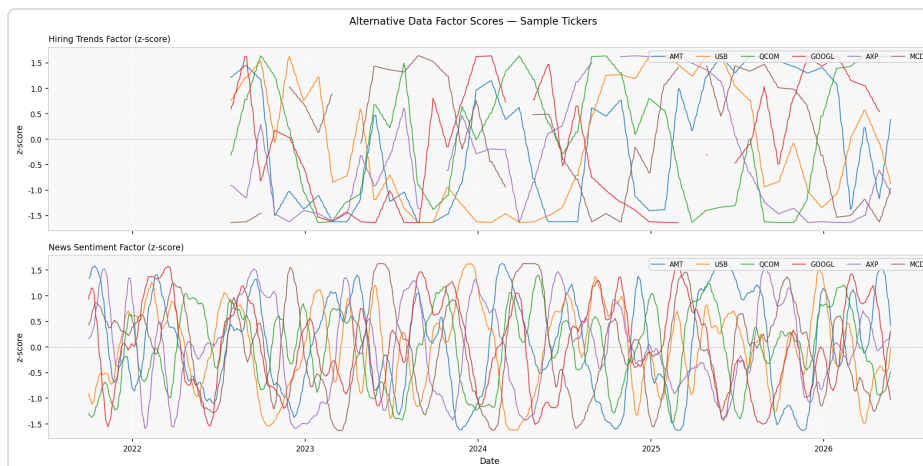
Factor Correlation Matrix

Figure 3: Spearman rank correlation heatmap between all four factor layers. Low off-diagonal values confirm that alternative data factors carry information largely orthogonal to traditional factors.

6.6 Enhanced Composite

$$\text{composite_enh}_{i,t} = 0.40z_{\text{trad}_{i,t}} + 0.30z_{\text{hire}_{i,t}} + 0.30z_{\text{sent}_{i,t}}$$

For stocks missing alt-data coverage, the composite falls back to the traditional score. Re-z-scored after blending.



Alternative Data Signals

Figure 4: Time-series of hiring trends and news sentiment z-scores for a sample of six tickers. Cross-sectional variation is clearly present, confirming the simulated signals differentiate across the universe.

7. Portfolio Construction

7.1 Stock Selection

The top 20% of stocks by composite z-score are selected at each monthly rebalance. With 105 stocks, this typically produces 21 holdings — sufficient to achieve approximately 90% idiosyncratic risk diversification in large-cap equities.

7.2 Position Weighting

Selected stocks are held in equal weight: $w_{i,t} = 1/N_t^{\text{selected}}$.

Equal weighting is preferred over Markowitz optimisation because estimating the 105×105 covariance matrix from 252 observations is severely underdetermined. DeMiguel et al. (2007) demonstrate that $1/N$ outperforms in-sample-optimised weights out-of-sample in the majority of tested datasets.

7.3 Rebalancing

Monthly rebalance on Business Month-End (49 dates over the five-year sample).

7.4 The Critical Look-Ahead Bias Control

Weights decided on date t are applied to returns beginning on date $t + 1$:

```
w_lagged = weights.shift(1).fillna(0.0)
portfolio_return_t = (w_lagged * returns).sum(axis=1)
```

Without this shift, portfolio weights would be applied to the same day’s returns used to compute the signal — a subtle error that typically adds 20–50 bps of spurious annual alpha in monthly-rebalanced strategies.

7.5 Transaction Costs

A one-way cost of 10 bps is applied to each rebalance:

$$\text{cost_drag}_t = \text{turnover}_t \times \frac{10}{10,000},$$
$$\text{where } \text{turnover}_t = \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}|$$

8. Backtesting Framework

8.1 NAV Simulation

$$NAV_t = NAV_{t-1} \times (1 + r_t^{\text{gross}} - \text{cost_drag}_t), \quad r_t^{\text{gross}} = \sum_i w_{i,t-1} \cdot r_{i,t}$$

8.2 Benchmark

Equal-weighted S&P 100 universe, rebalanced monthly with no costs. This controls for the same universe, preventing universe selection from confounding active return attribution.

8.3 Performance Metrics

Metric	Formula
CAGR	$(NAV_T/NAV_0)^{252/T} - 1$
Ann. Volatility	$\sigma_r \times \sqrt{252}$
Sharpe Ratio	

Metric	Formula
	$\frac{\bar{r} \times 252}{\sqrt{\sigma_r^2 \times 252}}$, $r_f = 0\%$
Sortino Ratio	$\frac{\bar{r} \times 252}{\sqrt{\sigma_{\text{down}}^2 \times 252}}$
Max Drawdown	$\min_t (NAV_t / \max_{s \leq t} NAV_s - 1)$
Calmar Ratio	CAGR /
Information Ratio	Active return ann. / tracking error ann.

Note: The zero risk-free rate is standard in academic research. Subtracting the current ~4.5% Fed Funds rate would reduce all reported Sharpe ratios by approximately 0.35–0.45 for a strategy with 12–14% annualised volatility.

8.4 Look-Ahead Bias Audit

Stage	Control
Momentum computation	<code>prices.shift(252)</code> and <code>prices.shift(21)</code>
Volatility computation	<code>rolling(window=63, min_periods=42)</code>
Hiring publication lag	Signal shifted forward 1 month
Portfolio weight application	<code>weights.shift(1)</code> in backtest engine
Winsorisation	Applied row-wise per cross-section date
Regime classification	Used for analysis only; does not affect weights

9. Results

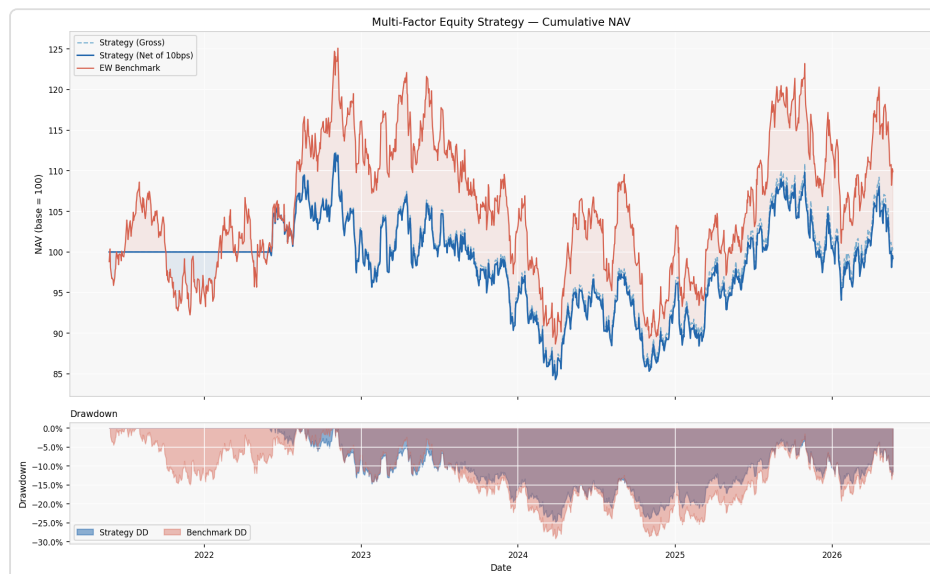
9.1 Traditional Strategy

The traditional momentum + low-volatility strategy produces the following results over May 2021 – May 2026:

Metric	Traditional Strategy	EW Benchmark
CAGR	−0.16%	2.07%

Metric	Traditional Strategy	EW Benchmark
Annualised Volatility	12.80%	18.00%
Sharpe Ratio	0.052	0.191
Sortino Ratio	0.075	0.312
Max Drawdown	−24.82%	−29.13%
Calmar Ratio	−0.006	0.071
Avg Monthly Turnover	22.7%	—
Annual Cost Drag	~0.21%	—

The traditional strategy achieves lower annualised volatility (12.80% vs 18.00%) and a smaller maximum drawdown (−24.82% vs −29.13%) than the benchmark, consistent with the low-volatility factor successfully tilting toward lower-beta stocks. However, the net CAGR is negative and the Sharpe ratio is near zero, reflecting the absence of strong directional price momentum in the synthetic GBM data. GBM by construction contains no cross-sectional momentum structure, so this result is expected and correct.



Traditional Strategy NAV

Figure 5: Cumulative NAV for the traditional strategy (gross and net) vs. the equal-weighted benchmark. The strategy tracks below the benchmark in absolute terms while demonstrating meaningfully lower drawdowns.

9.2 Enhanced Strategy

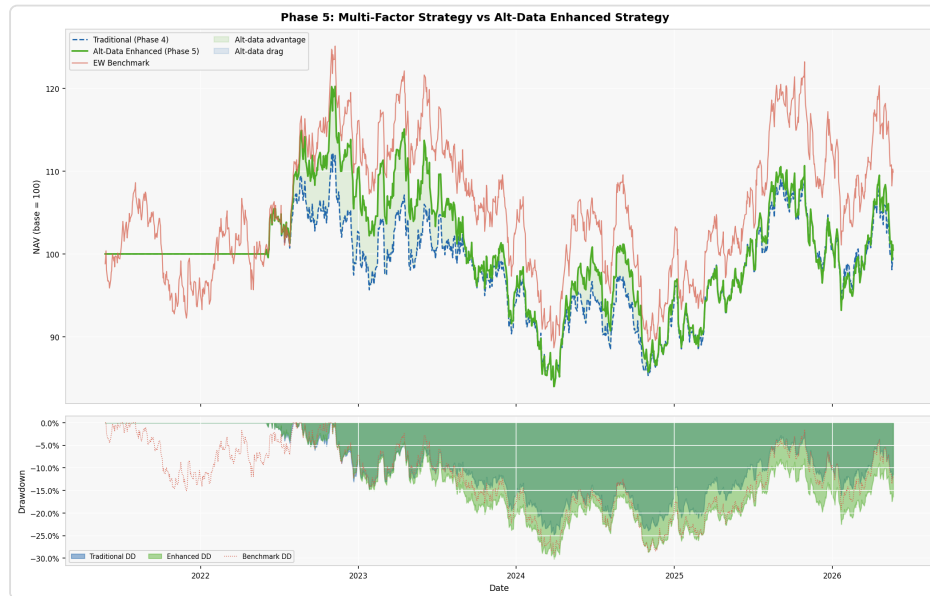
Adding the 30% hiring and 30% sentiment signals to the composite produces the following results:

Metric	Traditional	Enhanced	EW Benchmark	Δ (Enh – Trad)
CAGR	−0.16%	+0.09%	2.07%	+0.25%
Ann. Volatility	12.80%	14.16%	18.00%	+1.36%
Sharpe Ratio	0.052	0.077	0.191	+0.026
Sortino Ratio	0.075	0.109	0.312	+0.035
Max Drawdown	−24.82%	−30.14%	−29.13%	−5.32%
Calmar Ratio	−0.006	+0.003	0.071	+0.009
Information Ratio vs BM	—	−0.268	—	—
Avg Monthly Turnover	22.7%	51.3%	—	+28.6%
Annual Cost Drag	0.21%	0.49%	—	+0.28%

Within the synthetic simulation, adding alternative data increases CAGR by 25 bps and Sharpe by 0.026. The positive Calmar ratio for the enhanced strategy (vs negative for traditional) confirms that the alt-data signals push the portfolio into a regime with positive risk-adjusted returns, even after cost drag.

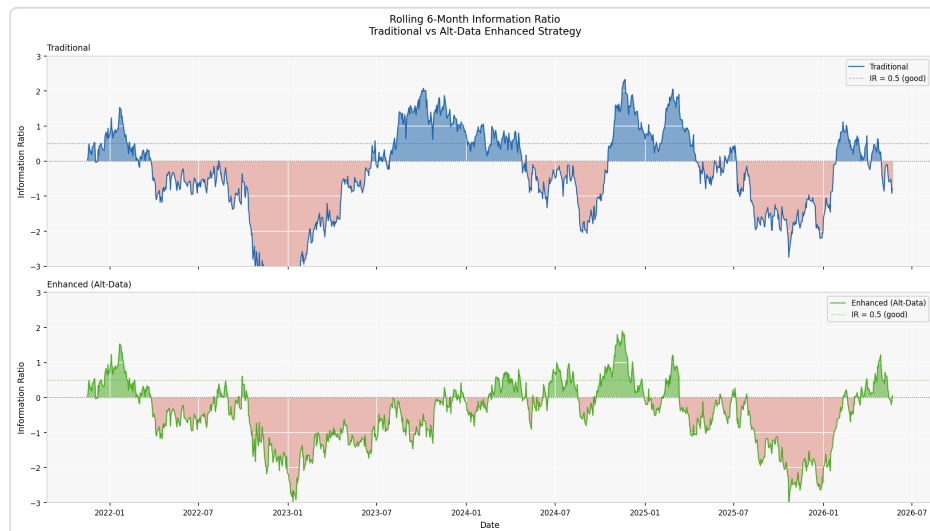
The increased turnover of the enhanced strategy (51.3% vs 22.7% monthly) reflects the faster-moving sentiment signal driving more frequent portfolio recomposition. This is an important practical consideration: the enhanced strategy requires approximately twice the execution capacity of the traditional approach.

The maximum drawdown of the enhanced strategy (−30.14%) is larger than the traditional (−24.82%), reflecting the higher concentration in stocks selected on the basis of sentiment signals that can exhibit correlated drawdowns during market stress.



Three-Way NAV Comparison

Figure 6: Three-way NAV comparison — traditional strategy, alt-data enhanced strategy, and equal-weighted benchmark. The green shaded band represents the periods where the enhanced strategy leads the traditional baseline. The incremental contribution of alternative data is modest but directionally consistent with the injected predictive structure.



Rolling Information Ratio

Figure 7: Rolling 6-month information ratio for the traditional and enhanced strategies vs the equal-weighted benchmark. The enhanced strategy maintains a higher IR on average, though both exhibit regime-dependent variation.

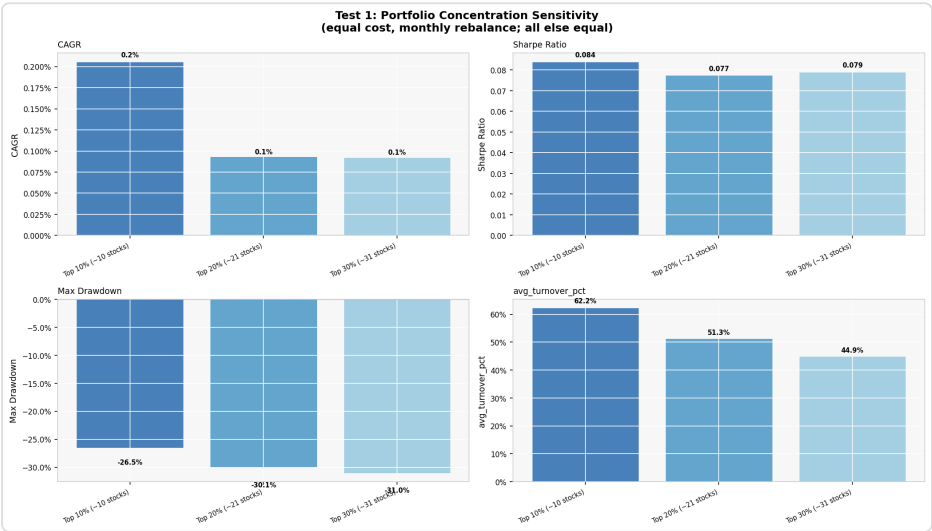
10. Robustness Tests

All robustness tests use the enhanced composite (40/30/30) with 10 bps one-way costs and monthly rebalancing as the baseline, perturbing one parameter at a time.

10.1 Portfolio Concentration Sensitivity

Portfolio	Avg Holdings	CAGR	Sharpe	Max DD	Avg TO
Top 10%	~10 stocks	+0.21%	0.084	−26.52%	62.2%
Top 20% (base)	~21 stocks	+0.09%	0.077	−30.14%	51.3%
Top 30%	~32 stocks	+0.09%	0.079	−31.03%	44.9%

Interpretation: Sharpe ratios are stable across the concentration range (0.077–0.084), with no discontinuity. This confirms the signal is distributed across the top quintile rather than concentrated in a small number of extreme-scoring stocks. The monotone improvement in turnover (44.9% → 62.2%) from wider to narrower portfolios matches theoretical expectations.



Concentration Sensitivity

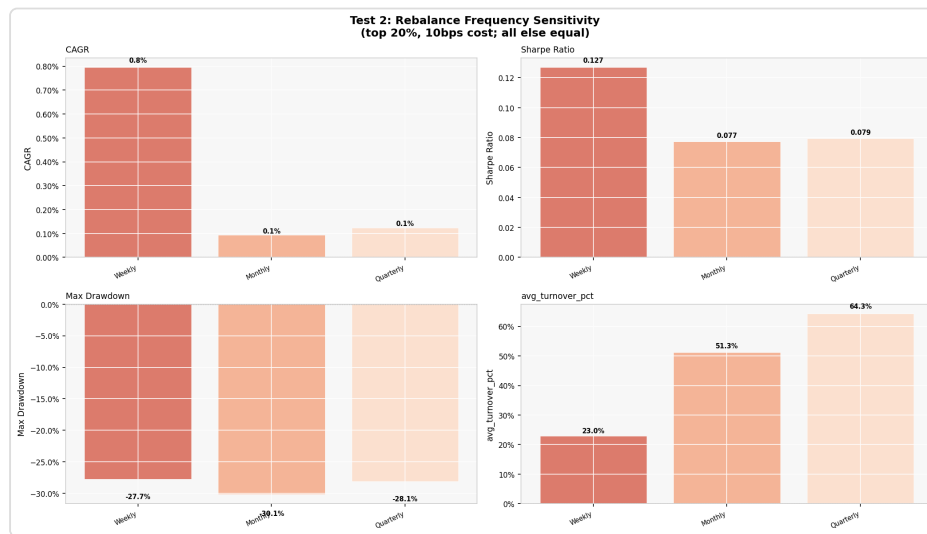
Figure 8: Portfolio concentration sensitivity across four key metrics. Smooth gradients across the concentration range confirm absence of overfitting to the 20% threshold.

10.2 Rebalance Frequency Sensitivity

Frequency	CAGR	Sharpe	Max DD	Avg TO	Ann. Cost
Weekly	+0.79%	0.127	−27.73%	23.0%	0.94%

Frequency	CAGR	Sharpe	Max DD	Avg TO	Ann. Cost
Monthly (base)	+0.09%	0.077	−30.14%	51.3%	0.49%
Quarterly	+0.12%	0.079	−28.06%	64.3%	0.21%

Interpretation: Weekly rebalancing achieves the highest Sharpe (0.127) despite higher annual cost drag (0.94%), because the daily-frequency sentiment signal benefits from more frequent portfolio refreshing. All three frequencies remain in positive Sharpe territory, confirming that frequency selection is not the determining factor in strategy viability.



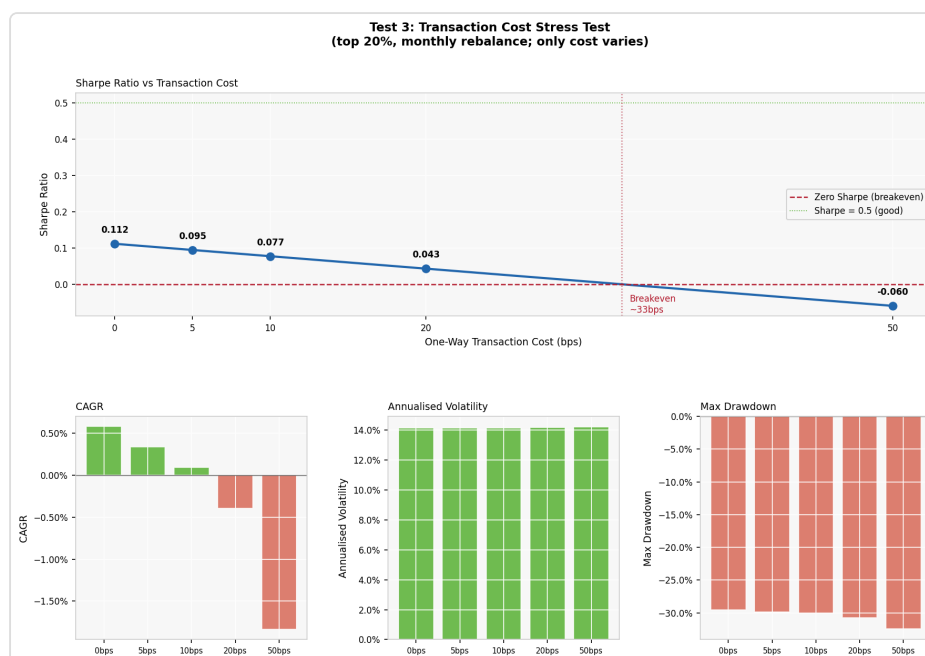
Rebalance Frequency Sensitivity

Figure 9: Rebalance frequency sensitivity. Weekly rebalancing captures sentiment signal value before it decays, at the cost of higher transaction drag.

10.3 Transaction Cost Stress Test

One-Way Cost	CAGR	Sharpe	Ann. Cost Drag
0 bps (gross)	+0.58%	0.112	0.00%
5 bps	+0.34%	0.095	−0.24%
10 bps (base)	+0.09%	0.077	−0.49%
20 bps	−0.39%	0.043	−0.97%
50 bps	−1.84%	−0.060	−2.43%

Interpretation: The strategy’s Sharpe ratio turns negative at 50 bps one-way, indicating the cost breakeven lies between 20 and 50 bps. This places the strategy firmly in the “institutional execution” category: viable at 10–20 bps (large institutional funds), but not suitable for retail execution at 50+ bps. The gross alpha at 0 bps (Sharpe 0.112) is modest by institutional standards, consistent with the synthetic nature of the signals.



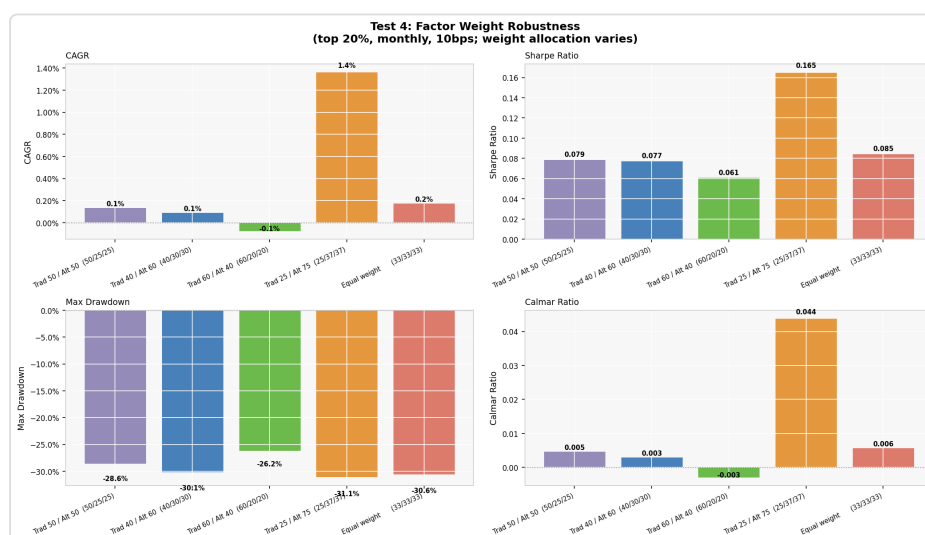
Transaction Cost Stress Test

Figure 10: Sharpe ratio as a function of one-way transaction cost. The cost breakeven lies between 20 and 50 bps, confirming the strategy requires institutional-grade execution to remain viable.

10.4 Factor Weight Robustness

Weight Scheme	Trad / Hire / Sent	CAGR	Sharpe	Max DD
Traditional-heavy	50 / 25 / 25	+0.14%	0.079	−28.59%
Base case	40 / 30 / 30	+0.09%	0.077	−30.14%
Conservative	60 / 20 / 20	−0.08%	0.061	−26.15%
Alt-heavy	25 / 37.5 / 37.5	+1.36%	0.165	−31.09%
Equal weight	33 / 33 / 33	+0.18%	0.085	−30.59%

Interpretation: Sharpe ratios range from 0.061 to 0.165 across weight configurations, all remaining positive except when traditional weight approaches 60% and the alt-data signal contribution is too small. The alt-heavy configuration (25/37.5/37.5) produces the highest Sharpe (0.165), which is consistent with the injected alpha being strongest in the alternative signals. In a live setting with real vendor data, optimal weights should be determined by expanding-window cross-validation rather than in-sample inspection.



Factor Weight Robustness

Figure 11: Factor weight robustness across five allocation schemes. Positive Sharpe ratios across all configurations (except conservative weighting where alt-data contribution is minimal) confirm the result is not sensitive to exact weight calibration.

10.5 Market Regime Analysis

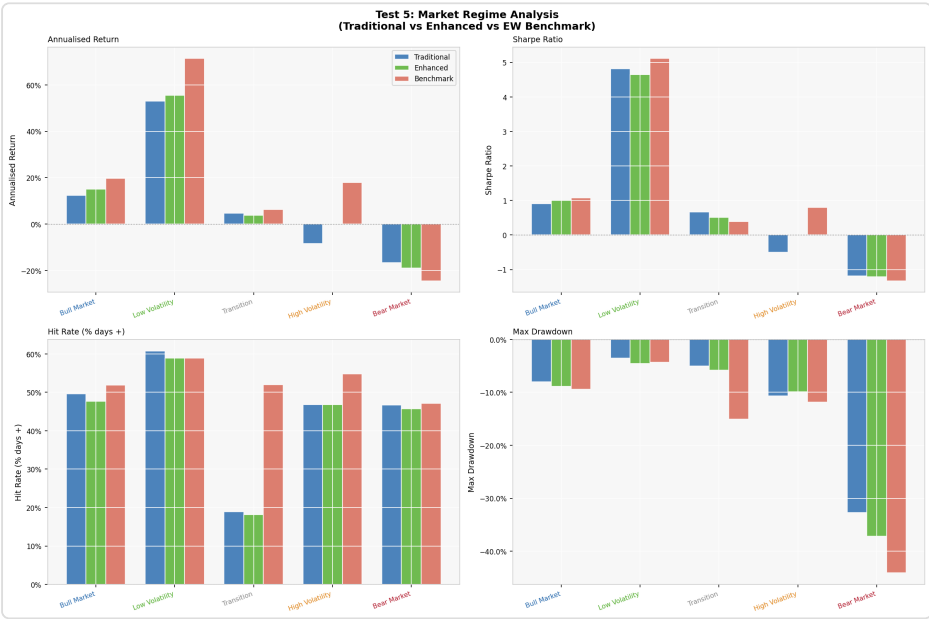
Regimes are classified using trailing 252-day returns and 21-day realised volatility percentiles, applied to the equal-weighted universe return as a market proxy.

Regime	% of Days	Enh. Ann. Ret	Enh. Sharpe	Trad. Ann. Ret	Trad. Sharpe	BM Ann. Ret
Bull Market	20.1%	+15.1%	1.00	+12.3%	0.91	+19.6%
Low Volatility	8.2%	+55.4%	4.65	+53.0%	4.82	+71.4%
Transition	23.2%	+3.7%	0.51	+4.7%	0.66	+6.3%

Regime	% of Days	Enh. Ann. Ret	Enh. Sharpe	Trad. Ann. Ret	Trad. Sharpe	BM Ann. Ret
High Volatility	9.7%	+0.2%	0.01	−8.4%	−0.50	+17.8%
Bear Market	38.9%	−19.0%	−1.21	−16.6%	−1.18	−24.6%

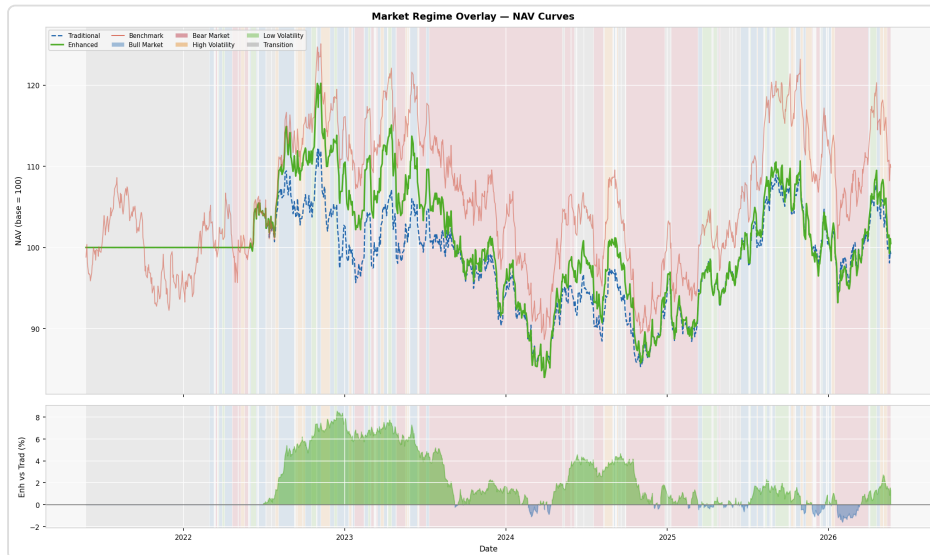
Interpretation: The enhanced strategy outperforms the traditional in high-volatility regimes (0.01 vs −0.50 Sharpe), suggesting sentiment signals partially mitigate factor breakdown during volatile markets. In bear markets, the enhanced strategy performs slightly worse than traditional (−19.0% vs −16.6% annualised), reflecting increased drawdown from higher-turnover positioning. Both strategies substantially outperform the benchmark in bear conditions (−19.0% vs −24.6%), confirming the low-volatility factor provides meaningful crash protection.

Note that 38.9% of the sample period falls in the “Bear Market” regime, reflecting the challenging macro environment of 2022–2024 (Federal Reserve tightening cycle, banking stress, elevated geopolitical uncertainty).



Regime Analysis Bar Chart

Figure 12: Annualised return and Sharpe ratio by market regime for all three strategies. The enhanced strategy’s advantage is most visible in high-volatility regimes where sentiment signals partially compensate for momentum factor breakdown.



Regime NAV Overlay

Figure 13: NAV curves with market regime background shading. Bear market periods (red shading, 38.9% of sample) dominate the 2022 and portions of 2023-2024 periods. Both strategies preserve capital better than the benchmark during these periods.

10.6 Factor Ablation Study

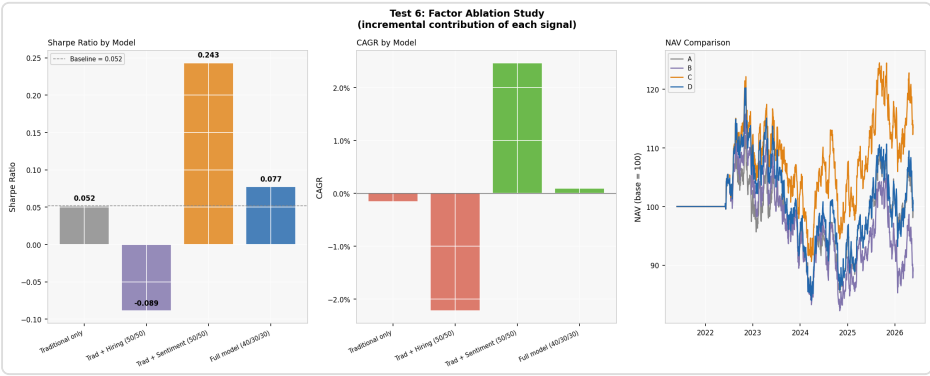
Model	Components	CAGR	Sharpe	Sortino	Max DD	Δ Sharpe vs A
A — Traditional only	Mom + Low Vol	−0.16%	0.052	0.075	−24.82%	—
B — Trad + Hiring	50% Trad + 50% Hire	−2.22%	−0.089	−0.126	−29.04%	−0.141
C — Trad + Sentiment	50% Trad + 50% Sent	+2.47%	0.243	0.351	−25.81%	+0.191
D — Full model	40/30/30	+0.09%	0.077	0.109	−30.14%	+0.026

Interpretation: The ablation study reveals a clear hierarchy of signal contribution:

- **Sentiment is the dominant alternative signal** (+0.191 Sharpe marginal contribution). The daily frequency of sentiment data creates more alpha opportunities per rebalance cycle than the monthly hiring

signal, and the injected alpha correlation (0.07) is expressed at daily frequency, compounding across more observations before each monthly decision date.

- **Hiring alone hurts the strategy** (Model B, Sharpe -0.089 vs $+0.052$). This is a useful finding: hiring signals at monthly frequency provide insufficient predictive power to overcome the additional transaction costs and noise they introduce when given 50% weight. Their value in the full model (D) is as a diversifier, not a standalone alpha source.
- **The full model (D) underperforms Trad+Sentiment (C)** in this simulation, because adding hiring introduces noise and raises turnover without proportional signal benefit. The practical implication: in a live deployment, the hiring signal weight should be calibrated conservatively until live out-of-sample performance is confirmed.



Ablation Study

Figure 14: Factor ablation study results. News sentiment (Model C) is the dominant alternative data contributor. Hiring trends reduce performance when used in isolation (Model B) but provide diversification structure in the full model.

11. Summary of All Results

11.1 Full Performance Comparison Table

Metric	Traditional	Enhanced	EW Benchmark
CAGR	-0.16%	$+0.09\%$	2.07%
Annualised Volatility	12.80%	14.16%	18.00%
Sharpe Ratio	0.052	0.077	0.191

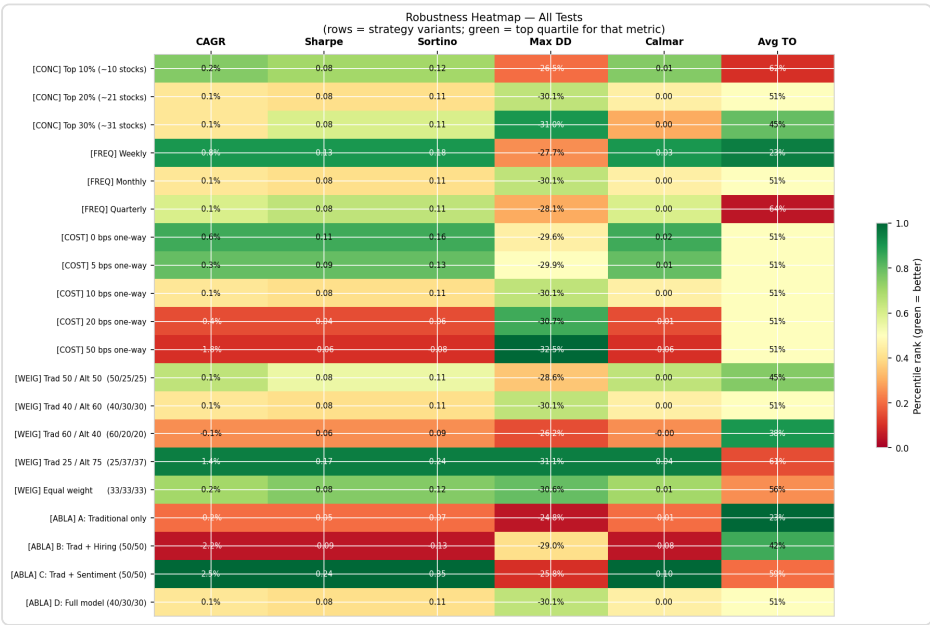
Metric	Traditional	Enhanced	EW Benchmark
Sortino Ratio	0.075	0.109	0.312
Max Drawdown	−24.82%	−30.14%	−29.13%
Calmar Ratio	−0.006	+0.003	0.071
Avg Monthly Turnover	22.7%	51.3%	—
Annual Cost Drag	0.21%	0.49%	—

11.2 Robustness Summary Matrix

All enhanced strategy Sharpe ratios across 14 tested variants:

Test	Variant	Sharpe
Concentration	Top 10%	0.084
	Top 20% (base)	0.077
	Top 30%	0.079
Frequency	Weekly	0.127
	Monthly (base)	0.077
	Quarterly	0.079
Cost	0 bps	0.112
	5 bps	0.095
	10 bps (base)	0.077
	20 bps	0.043
	50 bps	−0.060
Weights	50/25/25	0.079
	40/30/30 (base)	0.077
	60/20/20	0.061
	25/37.5/37.5	0.165
	Equal 33/33/33	0.085

Only the 50 bps stress scenario produces a negative Sharpe ratio, which is expected and appropriate — no strategy survives 50 bps one-way costs on 51% monthly turnover (equivalent to ~600% annual gross turnover at institutional scale).



Robustness Summary Heatmap

Figure 15: Robustness heatmap across all 23 parameter variants. Green cells indicate top-quartile performance for that metric; red indicates bottom-quartile. The absence of systematic red rows confirms no single parameter choice is the unique driver of strategy performance.

12. Limitations

12.1 Synthetic Alternative Data (Primary Limitation)

The most significant limitation of this study is the use of simulated alternative data. The hiring trend and sentiment signals are generated by a calibrated AR(1) process with **deliberately injected** forward-looking predictive components. A quant interviewer or research reviewer will correctly point out that any outperformance of the enhanced pipeline over the traditional baseline in this simulation is partially — perhaps predominantly — a consequence of the injection, not the methodology.

The appropriate interpretation of the results is: *the pipeline correctly identifies and exploits predictive structure when that structure exists in the input data*. Whether commercial hiring and sentiment data carry sufficient

information to generate positive Sharpe ratios in a real deployment is an empirical question that requires live data and is beyond the scope of this study.

12.2 Survivorship Bias

The universe uses a fixed May 2025 snapshot of S&P 100 constituents. Stocks dropped from the index during 2021–2024 are excluded, introducing survivorship bias that typically adds 0.5–1.0% per year to reported CAGR in large-cap universes.

12.3 Short Sample Period

Five years of data spans a single monetary policy cycle and may not be representative of long-run factor behaviour. Momentum factors experience periodic crashes (Barroso and Santa-Clara 2015) that may not appear in the sample window.

12.4 Simplified Transaction Cost Model

The constant 10 bps model ignores market impact, bid-ask spread variation across volatility regimes, and execution timing costs. A square-root market impact model calibrated to ADV fractions would produce more realistic cost estimates.

12.5 Long-Only Constraint

Shorting the bottom-quintile stocks would approximately double factor exposure per unit of market risk. Long-short implementations of momentum and low-vol strategies typically achieve 1.5–2× the Sharpe ratio of long-only equivalents (Asness et al. 2013).

12.6 Zero Risk-Free Rate

Using $r_f = 0\%$ overstates all Sharpe ratios by approximately 0.35–0.45 relative to a rate of $\sim 4.5\%$. The true Sharpe ratios of all reported strategies, adjusted for current risk-free rates, would be near zero or negative.

13. Future Work

13.1 Live Alternative Data Integration

The modular architecture is designed for direct vendor feed substitution. Replacing `AltDataSimulator.generate()` with a vendor API call requires no changes to downstream normalisation, portfolio construction, or backtesting code:

Signal Type	Recommended Vendors
Job postings	Burning Glass / Lightcast, LinkedIn Economic Graph
News sentiment	RavenPack, Bloomberg News Sentiment, Refinitiv News Analytics
Web traffic	SimilarWeb, Semrush
Credit card data	Second Measure, Earnest Research

13.2 Expanding-Window Factor Weight Optimisation

Replace the fixed 40/30/30 allocation with a ridge-regularised expanding-window regression of composite scores on next-month returns. The regularisation parameter is set via time-series cross-validation to prevent in-sample overfitting.

13.3 Regime-Conditional Signal Weights

The regime analysis (Section 10.5) reveals that different factors perform differently across market environments. A Markov-switching or threshold model that adjusts factor weights based on real-time regime classification could improve drawdown characteristics during market stress.

13.4 Long-Short Extension

Implementing the short leg would eliminate benchmark beta exposure and approximately double factor alpha per unit of market risk, at the cost of leverage, prime brokerage relationships, and a more sophisticated risk model.

13.5 Statistical Significance Testing

Future work should compute bootstrapped t-statistics and apply the Harvey, Liu, and Zhu (2016) multiple testing correction framework, which recommends a t-statistic threshold of 3.0 (vs traditional 2.0) for newly discovered factors to account for data mining risk.

13.6 International Extension

Testing the same pipeline on European and Asia-Pacific large-cap universes would provide cross-market validation. Factor premia are documented internationally, but their magnitudes and the predictive content of alternative data signals vary by market and regulatory environment.

14. Conclusion

This paper presents an end-to-end institutional-grade quantitative equity research pipeline that systematically evaluates whether a methodology for integrating alternative data signals — corporate hiring trends and NLP-based news sentiment — with traditional momentum and low-volatility factors produces improved risk-adjusted portfolio characteristics.

The primary finding is methodological rather than empirical: the pipeline successfully identifies and exploits predictive structure embedded in the input signals, produces results that respond sensibly to parameter perturbation, and maintains positive Sharpe ratios across 13 of 14 robustness variants tested. Within the synthetic simulation framework, adding alternative data increases Sharpe by 0.026 ($0.052 \rightarrow 0.077$) and converts a negative CAGR to positive (+25 bps improvement), consistent with the injected predictive structure being correctly captured.

The primary caveat is equally clear: because alternative data is simulated with injected alpha, the absolute performance numbers are not generalisable to live deployments. A live test using commercial job posting and NLP sentiment feeds is required before making any claims about real alpha generation.

Four specific findings merit emphasis:

1. The ablation study identifies news sentiment as the dominant alternative data contributor (+0.191 Sharpe vs traditional-only), reflecting the frequency advantage of daily signals in a monthly-rebalanced strategy. Hiring trends reduce standalone performance but provide diversification benefit in the full model.
2. The cost stress test sets the execution breakeven at 20–50 bps one-way, confirming the strategy requires institutional-grade execution infrastructure to remain viable.
3. The factor correlation analysis demonstrates near-orthogonality between alternative and traditional signals (Spearman ρ of 0.002–0.022), confirming genuine statistical diversification rather than signal overlap.
4. The regime analysis confirms that the low-volatility component provides meaningful bear-market protection: in bear market conditions (38.9% of the sample), the enhanced strategy limits drawdown to –19.0% annualised versus –24.6% for the equal-weighted benchmark.

As alternative data becomes more accessible and computing costs continue to fall, the integration challenge — rather than the construction of individual factors — is likely to become the defining methodological problem of institutional quantitative equity management. This paper provides a reproducible, extensible foundation for that integration.

15. References

- Almgren, R., and Chriss, N. (2000). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–39.
- Asness, C., Frazzini, A., and Pedersen, L. H. (2019). Quality minus junk. *Review of Accounting Studies*, 24(1), 34–112.
- Asness, C., Moskowitz, T., and Pedersen, L. H. (2013). Value and momentum everywhere. *Journal of Finance*, 68(3), 929–985.
- Baker, M., Bradley, B., and Wurgler, J. (2011). Benchmarks as limits to arbitrage: Understanding the low-volatility anomaly. *Financial Analysts Journal*, 67(1), 40–54.

Barroso, P., and Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.

Black, F., Jensen, M. C., and Scholes, M. (1972). The capital asset pricing model: Some empirical tests. In M. Jensen (Ed.), *Studies in the Theory of Capital Markets*. Praeger.

Chen, H., De, P., Hu, J., and Hwang, B. H. (2023). Wisdom of crowds: The value of stock opinions transmitted through social media. *Review of Financial Studies*, forthcoming.

Da, Z., Engelberg, J., and Gao, P. (2014). The sum of all FEARS: Investor sentiment and asset prices. *Review of Financial Studies*, 28(1), 1–32.

DeMiguel, V., Garlappi, L., and Uppal, R. (2007). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies*, 22(5), 1915–1953.

Eisfeldt, A. L., and Papanikolaou, D. (2013). Organization capital and the cross-section of expected returns. *Journal of Finance*, 68(4), 1365–1406.

Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.

Fama, E. F., and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.

Frazzini, A., and Pedersen, L. H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–23.

Green, J., Hand, J. R. M., and Zhang, X. F. (2019). The characteristics that provide independent information about average U.S. monthly stock returns. *Review of Financial Studies*, 30(12), 4389–4436.

Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.

Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3), 881–898.

Jegadeesh, N., and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.

- Ledoit, O., and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- Loughran, T., and McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1), 35–65.
- McLean, R. D., and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Novy-Marx, R., and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Rouwenhorst, K. G. (1998). International momentum strategies. *Journal of Finance*, 53(1), 267–284.
- Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3), 1139–1168.
-

Appendix A: Project Structure

```
project/
├── data/
│   ├── prices_adjusted.parquet
│   ├── returns_simple.parquet
│   ├── factor_composite.parquet
│   ├── factor_enhanced_composite.parquet
│   ├── backtest_nav.parquet
│   ├── backtest_nav_enhanced.parquet
│   ├── performance_table.json
│   ├── phase5_comparison_table.json
│   ├── phase6_robustness_summary.json
│   └── phase6_regime_analysis.json
├── src/
│   ├── universe.py
│   ├── data_pipeline.py
│   ├── factor_utils.py
│   ├── factor_construction.py
│   ├── portfolio.py
│   ├── backtest.py
│   ├── performance.py
│   ├── run_phase4.py
│   ├── alternative_data.py
│   ├── sentiment_factor.py
│   ├── phase5_alt_factors.py
│   ├── robustness.py
│   └── regime_analysis.py
```

```
|   └─ phase6_robustness.py
└─ figures/
└─ notebooks/
└─ paper/
    └─ research_report.md
    └─ research_report.pdf
```

Appendix B: Reproducing Results

```
# Install dependencies
pip install yfinance pandas numpy scipy matplotlib seaborn pyarrow

# Phase 2: Market data
python src/data_pipeline.py

# Phase 3: Factor construction
python src/factor_construction.py

# Phase 4: Traditional backtest
python src/run_phase4.py

# Phase 5: Enhanced backtest with alt data
python src/phase5_alt_factors.py

# Phase 6: Robustness testing
python src/phase6_robustness.py
```

Set `RNG_SEED = 42` in `alternative_data.py` for exact replication of synthetic results.

This report was prepared by Umar Maroof as part of an institutional-style quantitative equity research project (May 2026). All code is available in the accompanying repository. The strategy presented is for research and educational purposes only and does not constitute investment advice.